

## TROMSO AP, Norway

WMO: 010250

Lat: 69.683N

Lon: 18.919E

Elev: 31

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 9.5        | 12.1 | 2.9                           | 6.4 | 13.1 | 5.4 | 7.3 | 14.4 | 34.7                  | 35.4 | 31.2 | 33.2 | 4.3                   | 120  | 0.783 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 68.5            | 58.2 | 64.9 | 56.2 | 61.7 | 54.8 | 59.3                | 66.6 | 57.3 | 63.5 | 55.6 | 60.6 | 8.1                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 55.8                            | 66.3 | 61.8 | 54.0 | 62.1 | 59.5 | 52.6 | 58.9 | 57.9 | 26.0          | 66.4 | 24.6 | 63.7 | 23.6 | 60.7 | 70.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 29.4          | 25.6 | 22.6 | 5.6                        | 74.5 | 2.8                | 4.3 | 3.7                                                | 77.6 | 2.0        | 80.2 | 0.5        | 82.6 | -1.5       | 85.7 |     |     |
| (5)               |      |      | 4.7                        | 62.8 | 2.8                | 3.2 | 2.7                                                | 65.1 | 1.1        | 67.0 | -0.4       | 68.8 | -2.5       | 71.1 | (4) | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov  | Dec  |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)  | (p)  |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 38.1   | 26.4  | 25.8  | 28.4  | 33.9  | 41.9  | 48.6  | 53.9  | 52.2  | 46.3  | 38.0 | 32.1 | 29.0 | (6)  |
| (7)  |                                                                                  | DBStd       | 11.56  | 7.28  | 7.38  | 6.37  | 5.59  | 5.76  | 5.27  | 5.13  | 4.57  | 5.17  | 6.22 | 6.22 | 7.19 | (7)  |
| (8)  |                                                                                  | HDD50       | 4648   | 732   | 678   | 671   | 484   | 259   | 88    | 16    | 26    | 132   | 374  | 537  | 651  | (8)  |
| (9)  |                                                                                  | HDD65       | 9816   | 1197  | 1098  | 1136  | 933   | 715   | 493   | 346   | 396   | 561   | 838  | 987  | 1116 | (9)  |
| (10) |                                                                                  | CDD50       | 308    | 0     | 0     | 0     | 0     | 9     | 45    | 137   | 95    | 21    | 1    | 0    | 0    | (10) |
| (11) |                                                                                  | CDD65       | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0    | 0    | 0    | (11) |
| (12) |                                                                                  | CDH74       | 11     | 0     | 0     | 0     | 0     | 0     | 1     | 8     | 2     | 0     | 0    | 0    | 0    | (12) |
| (13) |                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | 0    | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 9.7    | 11.7  | 11.7  | 11.1  | 8.9   | 9.0   | 9.2   | 8.1   | 7.5   | 7.9   | 9.3  | 10.4 | 11.6 | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 39.4   | 3.9   | 3.5   | 3.2   | 2.4   | 2.0   | 2.2   | 2.6   | 3.0   | 3.9   | 5.2  | 3.8  | 4.3  | (15) |
| (16) |                                                                                  | PrecMax     | 62.8   | 13.4  | 9.8   | 7.2   | 7.3   | 5.8   | 6.4   | 6.4   | 7.6   | 9.2   | 13.1 | 9.5  | 10.4 | (16) |
| (17) |                                                                                  | PrecMin     | 25.3   | 0.0   | 0.1   | 0.3   | 0.2   | 0.2   | 0.2   | 0.2   | 0.6   | 0.2   | 0.7  | 0.4  | 0.1  | (17) |
| (18) |                                                                                  | PrecStd     | 8.0    | 2.4   | 1.9   | 1.8   | 1.4   | 1.2   | 1.3   | 1.3   | 1.7   | 2.1   | 2.5  | 2.0  | 2.2  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 42.7  | 42.0  | 43.3  | 52.4  | 65.0  | 70.5  | 74.4  | 71.7  | 62.6  | 55.0 | 48.4 | 44.9 | (19) |
| (20) |                                                                                  |             | MCWB   | 38.9  | 37.9  | 37.5  | 44.2  | 52.7  | 57.7  | 62.2  | 59.8  | 54.4  | 48.9 | 44.7 | 40.9 | (20) |
| (21) |                                                                                  | 2%          | DB     | 40.6  | 39.9  | 41.0  | 47.8  | 58.5  | 64.7  | 69.8  | 66.5  | 59.0  | 51.3 | 45.8 | 42.7 | (21) |
| (22) |                                                                                  |             | MCWB   | 37.6  | 36.5  | 37.0  | 41.8  | 48.6  | 54.3  | 59.5  | 57.4  | 53.2  | 46.8 | 42.5 | 39.7 | (22) |
| (23) |                                                                                  | 5%          | DB     | 38.7  | 38.0  | 39.1  | 44.1  | 54.2  | 60.7  | 66.2  | 62.6  | 56.2  | 49.0 | 43.2 | 40.8 | (23) |
| (24) |                                                                                  |             | MCWB   | 36.0  | 35.3  | 35.6  | 39.6  | 46.2  | 52.6  | 57.6  | 55.9  | 51.1  | 45.5 | 39.8 | 37.5 | (24) |
| (25) |                                                                                  | 10%         | DB     | 36.7  | 35.9  | 37.4  | 41.8  | 50.7  | 56.9  | 62.7  | 59.5  | 53.8  | 46.8 | 41.2 | 38.9 | (25) |
| (26) |                                                                                  |             | MCWB   | 34.2  | 33.4  | 34.2  | 37.7  | 44.4  | 50.6  | 56.0  | 54.4  | 49.9  | 43.9 | 38.1 | 35.7 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 39.8  | 38.9  | 38.9  | 44.9  | 53.5  | 58.5  | 63.5  | 61.7  | 56.1  | 49.8 | 45.6 | 41.9 | (27) |
| (28) |                                                                                  |             | MCD B  | 41.2  | 41.0  | 41.2  | 50.1  | 64.4  | 68.3  | 72.8  | 68.5  | 60.8  | 53.3 | 47.7 | 43.4 | (28) |
| (29) |                                                                                  | 2%          | WB     | 37.8  | 37.0  | 37.5  | 42.3  | 49.5  | 55.4  | 60.3  | 58.4  | 53.6  | 47.6 | 42.7 | 40.0 | (29) |
| (30) |                                                                                  |             | MCD B  | 39.5  | 39.1  | 40.1  | 46.4  | 57.6  | 62.8  | 68.0  | 64.5  | 57.5  | 50.6 | 45.0 | 42.3 | (30) |
| (31) |                                                                                  | 5%          | WB     | 36.2  | 35.7  | 36.0  | 40.1  | 47.0  | 53.1  | 58.3  | 56.6  | 52.0  | 45.8 | 40.3 | 37.7 | (31) |
| (32) |                                                                                  |             | MCD B  | 38.3  | 37.8  | 38.7  | 43.5  | 52.9  | 59.2  | 65.1  | 61.7  | 55.5  | 48.4 | 42.8 | 40.2 | (32) |
| (33) |                                                                                  | 10%         | WB     | 34.2  | 33.7  | 34.4  | 38.2  | 44.9  | 51.1  | 56.4  | 54.9  | 50.4  | 44.2 | 38.2 | 35.8 | (33) |
| (34) |                                                                                  |             | MCD B  | 36.3  | 35.6  | 37.0  | 41.3  | 49.9  | 56.2  | 61.8  | 59.1  | 53.4  | 46.6 | 40.7 | 38.3 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.9   | 8.2   | 8.7   | 9.3   | 9.7   | 9.3   | 10.3  | 10.1  | 9.4   | 7.7  | 7.4  | 7.9  | (35) |
| (36) |                                                                                  |             | MCD BR | 7.7   | 8.0   | 7.4   | 11.5  | 16.3  | 15.9  | 17.4  | 16.7  | 13.3  | 8.7  | 8.6  | 8.1  | (36) |
| (37) |                                                                                  |             | MCWBR  | 6.6   | 6.7   | 5.8   | 7.5   | 8.9   | 8.3   | 9.1   | 9.2   | 8.0   | 6.6  | 7.1  | 6.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 7.6   | 8.0   | 7.1   | 10.5  | 15.1  | 14.6  | 16.2  | 14.8  | 12.2  | 8.0  | 8.2  | 8.0  | (38) |
| (39) |                                                                                  |             | MCWBR  | 6.9   | 7.2   | 5.9   | 7.3   | 8.6   | 8.3   | 9.1   | 8.9   | 7.9   | 6.8  | 7.1  | 7.2  | (39) |
| (40) |                                                                                  |             |        |       |       |       |       |       |       |       |       |       |      |      |      |      |
| (41) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | N/A    | 0.276 | 0.324 | 0.341 | 0.352 | 0.350 | 0.354 | 0.342 | 0.322 | 0.283 | N/A  | N/A  | (40) |      |
| (42) |                                                                                  | taud        | N/A    | 2.046 | 2.098 | 2.210 | 2.365 | 2.452 | 2.476 | 2.495 | 2.499 | 2.192 | N/A  | N/A  | (41) |      |
| (43) |                                                                                  | Ebn at Noon | N/A    | 135   | 208   | 249   | 263   | 268   | 262   | 252   | 223   | 135   | 0    | 0    | (42) |      |
|      |                                                                                  | Edh at Noon | N/A    | 14    | 26    | 32    | 31    | 30    | 28    | 24    | 18    | 11    | 0    | 0    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 3      | 83    | 426   | 1004  | 1453  | 1525  | 1379  | 979   | 517   | 158   | 0    | 0    | (44) |      |
| (45) |                                                                                  | RadStd      | 0      | 17    | 58    | 97    | 97    | 195   | 143   | 80    | 60    | 24    | 0    | 0    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | -1.08 | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (22 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page